

OC Core Methods and Reproducibility Companion v1.3.3

Version: 1.3.3 Concept DOI for PDF citation: 10.5281/zenodo.17899134 Author: Alexander Yashin

Abstract

This review-space artifact is assembled from the current OC Core release aggregator, science-to-structure mapping, package cascade, terminal text contracts, and deterministic transition rules. It is not a GitHub or Zenodo publication action.

Reader Contract

Read this document as an assembled scientific route. Claims are bounded by their proof, evidence, replay, comparator, or falsifier route; unsupported all-domain or superiority claims are not promoted by package assembly.

Block 5

Define Claim Classes

Define Claim Classes is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Claim Classes. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Claim Classes to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Claim Classes states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Claim Classes synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary

remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Formal Proof Standard

Define Formal Proof Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Formal Proof Standard. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Formal Proof Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Formal Proof Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Formal Proof Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Executable Semantics Standard

Define Executable Semantics Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Executable Semantics Standard. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Executable Semantics Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Executable Semantics Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Executable Semantics Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Empirical Evidence Standard

Define Empirical Evidence Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Empirical Evidence Standard. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Empirical Evidence Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Empirical Evidence Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must

route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Empirical Evidence Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Simulation

Define Simulation and Counterexample Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Simulation and Counterexample Standard. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Simulation and Counterexample Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Simulation and Counterexample Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Simulation and Counterexample Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Negative Controls

Define Negative Controls and Falsifiers is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Negative Controls and Falsifiers. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Negative Controls and Falsifiers to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Negative Controls and Falsifiers states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Negative Controls and Falsifiers synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Source, Artifact,

Define Source, Artifact, and Traceability Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Source, Artifact, and Traceability Standard. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Source, Artifact, and Traceability Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route

uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Source, Artifact, and Traceability Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Source, Artifact, and Traceability Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Editorial

Define Editorial and Reviewer Evidence Standard is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Editorial and Reviewer Evidence Standard. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Editorial and Reviewer Evidence Standard to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Editorial and Reviewer Evidence Standard states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Editorial and Reviewer Evidence Standard synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Block 10

Define Formalization Strategy

Define Formalization Strategy is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Formalization Strategy. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Formalization Strategy to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Formalization Strategy states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Formalization Strategy synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Lean/Lake Subset

Define Lean/Lake Subset is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model

route for Lean/Lake Subset. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Lean/Lake Subset to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Lean/Lake Subset states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Lean/Lake Subset synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Typed Structures in Lean

Define Typed Structures in Lean is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Typed Structures in Lean. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Typed Structures in Lean to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only

becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Typed Structures in Lean states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Typed Structures in Lean synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Machine-Checked Theorem Subset

Define Machine-Checked Theorem Subset is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Machine-Checked Theorem Subset. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Machine-Checked Theorem Subset to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Machine-Checked Theorem Subset states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Machine-Checked Theorem Subset synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already

named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Finite Model Semantics

Define Finite Model Semantics is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Finite Model Semantics. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Finite Model Semantics to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Finite Model Semantics states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Finite Model Semantics synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Witness Cases

Define Witness Cases is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Witness Cases. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Witness Cases to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Witness Cases states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Witness Cases synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Tamper

Define Tamper and Negative Controls is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Tamper and Negative Controls. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Tamper and Negative Controls to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Tamper and Negative Controls states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must

route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Tamper and Negative Controls synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Limits of the Formalized Subset

Define Limits of the Formalized Subset is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Limits of the Formalized Subset. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Limits of the Formalized Subset to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Limits of the Formalized Subset states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Limits of the Formalized Subset synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Block 11

Define Evidence Ledger

Define Evidence Ledger is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Evidence Ledger. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Evidence Ledger to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Evidence Ledger states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Evidence Ledger synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Mathematics Anchor

Define Mathematics Anchor is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Mathematics Anchor. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Mathematics Anchor to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite

semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Mathematics Anchor states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Mathematics Anchor synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Physics Evidence Lane

Define Physics Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Physics Evidence Lane. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Physics Evidence Lane to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Physics Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Physics Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Chemistry Evidence Lane

Define Chemistry Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Chemistry Evidence Lane. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Chemistry Evidence Lane to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Chemistry Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Chemistry Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Biology Evidence Lane

Define Biology Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Biology Evidence Lane. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary

before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Biology Evidence Lane to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Biology Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Biology Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Systems

Define Systems and Civilizational Evidence Lane is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Systems and Civilizational Evidence Lane. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Systems and Civilizational Evidence Lane to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Systems and Civilizational Evidence Lane states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Systems and Civilizational Evidence Lane synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Simulations

Define Simulations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition `model route` for Simulations. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Simulations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Simulations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Simulations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Adversarial Simulations

Define Adversarial Simulations is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Adversarial Simulations. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Adversarial Simulations to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Adversarial Simulations states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Adversarial Simulations synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Target-Blind

Define Target-Blind and Held-Out Evidence is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Target-Blind and Held-Out Evidence. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Target-Blind and Held-Out Evidence to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route

uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Target-Blind and Held-Out Evidence states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Target-Blind and Held-Out Evidence synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Evidence Promotion Boundaries

Define Evidence Promotion Boundaries is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Evidence Promotion Boundaries. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Evidence Promotion Boundaries to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Evidence Promotion Boundaries states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Evidence Promotion Boundaries synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Block 17

Define Build Environment

Define Build Environment is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Build Environment. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Build Environment to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Build Environment states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Build Environment synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Data Availability

Define Data Availability is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model

route for Data Availability. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Data Availability to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Data Availability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Data Availability synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Code Availability

Define Code Availability is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition route for Code Availability. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Code Availability to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data;

otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Code Availability states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Code Availability synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Artifact Inventory

Define Artifact Inventory is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Artifact Inventory. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Artifact Inventory to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Artifact Inventory states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Artifact Inventory synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack

response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Checksums

Define Checksums and Integrity is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Checksums and Integrity. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Checksums and Integrity to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Checksums and Integrity states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Checksums and Integrity synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Replay Protocols

Define Replay Protocols is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Replay Protocols. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence

slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Replay Protocols to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Replay Protocols states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Replay Protocols synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Source-to-PDF Trace

Define Source-to-PDF Trace is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Source-to-PDF Trace. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Source-to-PDF Trace to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Source-to-PDF Trace states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission

to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Source-to-PDF Trace synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Reproducibility Limits

Define Reproducibility Limits is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Reproducibility Limits. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Reproducibility Limits to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Reproducibility Limits states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Reproducibility Limits synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Block 18

Define Versioning

Define Versioning and Release Identity is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Versioning and Release Identity. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Versioning and Release Identity to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Versioning and Release Identity states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Versioning and Release Identity synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Public Release Asset Set

Define Public Release Asset Set is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Public Release Asset Set. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Public Release Asset Set to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite

semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Public Release Asset Set states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Public Release Asset Set synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Journal Package Map

Define Journal Package Map is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Journal Package Map. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Journal Package Map to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Journal Package Map states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Journal Package Map synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Citation

Define Citation and Metadata Governance is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Citation and Metadata Governance. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Citation and Metadata Governance to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Citation and Metadata Governance states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Citation and Metadata Governance synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Publication Boundary

Define Publication Boundary is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the `definition_model` route for Publication Boundary. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary

before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Publication Boundary to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Publication Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Publication Boundary synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Define Owner Approval Boundary

Define Owner Approval Boundary is introduced here as a reader-facing obligation, not as a file name or process label. The reader task is to fill one future paragraph that completes the definition_model route for Owner Approval Boundary. The paragraph draws on formal model and foundation, prior art novelty and comparator, didactic synthesis and reader guidance and states the local vocabulary before any proof, replay, or comparison is allowed to carry weight. Its boundary is conservative: may define model terms and scope; may not promote empirical or universal claims without later proof/evidence slots. With the object named, the next move is to expose the support route instead of relying on assertion.

Bind Owner Approval Boundary to proof, evidence, or replay carries the evidential burden for the surrounding claim. The release text must show how the reader moves from prose to proof, finite semantics, replay material, comparator evidence, or source trace. The mapped route uses formal model and foundation, machine checked and finite semantics, empirical computational evidence; exact paths and hashes stay in the source trace so the paragraph remains readable. The support is promoted only as far as this boundary permits: may support promoted claims only through cited proof/data; otherwise marks the claim as partial, planned, or missing. The support route only becomes reviewable when its boundary, falsifier, and negative side are visible.

State limits and falsifiers for Owner Approval Boundary states what would make the local claim weaker, false, incomplete, or research-only. The release cannot use the presence of evidence as permission to overstate scope. It must name the falsifier, negative control, reviewer objection, or residual risk carried by review attack response and limits, empirical computational evidence, prior art novelty and comparator. The governing boundary is: must prevent overclaiming and must route unsupported strength to limits or background research. Once the boundary is explicit, the manuscript may synthesize the local consequence without inflating it.

Synthesize Owner Approval Boundary synthesizes the local route for the reader. It states what has been established, what remains bounded, and why the next section follows. The synthesis draws on didactic synthesis and reader guidance, reproducibility governance and artifacts, review attack response and limits but does not add new scientific strength beyond the evidence already named. The boundary remains: may synthesize established local results but must not add new unsupported claims. The next obligation begins by defining the next object before asking the reader to accept claims about it.

Source Trace

Exact source bindings, path hashes, quality scorer hooks, and transition records are recorded in the generated artifact package manifest. They are kept out of the main prose to avoid turning the document into a ledger dump.